Nanotechnology in Medicine

Speech for the presentation, slide by slide

Slide 1 - Nanotechnology in Medicine

Welcome. A nanometer is one billionth of a meter, and at this scale we can build tools that diagnose, treat and repair the body. Let's see how.

Slide 2 - Why the nanoscale matters

Biology works at the nanoscale: DNA is about 2 nm wide, proteins 2 to 10 nm, viruses 20 to 200 nm. Nanoparticles match this size, cross barriers larger objects cannot, and offer a large surface to carry drugs.

Slide 3 - The nanocarrier toolbox

Liposomes are fat bubbles that hold drugs. Lipid nanoparticles protect RNA. Polymer particles release drugs slowly. Gold and iron-oxide particles serve for imaging and heating. Dendrimers and quantum dots carry drugs and label cells.

Slide 4 - Targeted drug delivery

The aim is to deliver the drug only where it is needed. Particles accumulate in tumors through leaky vessels, or bind diseased cells through surface antibodies. Some release their drug on a trigger such as acidity, heat or light. Doxil, a liposomal chemotherapy, reduces heart toxicity.

Slide 5 - mRNA vaccines

Messenger RNA breaks down quickly, so lipid nanoparticles of 80 to 100 nm wrap and deliver it. Our cells make a harmless protein and the immune system learns to recognize it. This is how the COVID-19 vaccines work.

Slide 6 - Fighting cancer

Gold nanoshells heat tumors under near-infrared light. Iron-oxide particles heat them under a magnetic field. Abraxane treats breast and pancreatic cancer, and hafnium oxide particles boost radiotherapy. Healthy tissue is spared more than with classic chemotherapy.

Slide 7 - Nano-diagnostics and imaging

Iron-oxide particles improve MRI contrast. Quantum dots label cells brightly. Gold nanoparticles create the line in pregnancy tests. Biosensors detect a few cancer-marker molecules in one drop of blood. Earlier detection means better survival.

Slide 8 - Regenerative medicine

Nano-scaffolds guide cell growth like natural tissue. Nano-hydroxyapatite is used in bone grafts and dental implants. Nanofibers help nerves regrow, silver dressings protect wounds, and particles release growth factors slowly.

Slide 9 - Fighting infection and the road ahead

Silver and zinc-oxide particles attack antibiotic-resistant bacteria. Theranostic particles diagnose and treat at once. DNA origami robots open on contact with target cells, and new carriers aim to cross the blood-brain barrier.

Slide 10 - Key milestones

1959: Feynman's famous talk. 1995: Doxil, the first nanomedicine. 2005: Abraxane. 2018: Onpattro, the first lipid-nanoparticle RNA drug. 2020: mRNA COVID-19 vaccines.

Slide 11 - Advantages and disadvantages

Advantages: precise delivery, fewer side effects, lower doses, earlier diagnosis, protection of fragile drugs, combined imaging and therapy, and tissue repair. Disadvantages: unclear long-term toxicity, buildup in the liver or spleen, hard and costly manufacturing, complex regulation, and ethical and privacy concerns.

Slide 12 - Thank you

Nanomedicine offers smaller tools and bigger hope for patients, provided we develop it safely. Thank you. Any questions?